

2025

(Session : 2024-26)

ESUE(I) — MBA

(CC – 107)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Candidates are permitted to use Simple Calculator.

Group – A (Compulsory)

1. Choose the correct answer of the following :

1×5 = 5

(a) The value of coefficient of correlation varies from :

- i. 0 to 1
- ii. 1 to 100
- iii. -1 to +1
- iv. 0 to 100

(b) In a Poisson Distribution, the arithmetic mean is :

- i. Equal to variance
- ii. Greater than variance
- iii. Less than variance
- iv. Half of variance

(c) In a Positively Skewed Distribution :

- i. Mean is maximum
- ii. Median is maximum
- iii. Mode is maximum
- iv. Mean, Median and Mode are equal

(d) ANOVA (Analysis of variance) can be used for :

- i. One population
- ii. Two populations

- iii. One sample
- iv. Three or more populations

(e) When a coin is tossed thrice; the number of outcomes are :

- i. 2
- ii. 4
- iii. 6
- iv. 8

2. Briefly explain the concept of Statistical Quality Control. What is a Control Chart? What are the types of Control Charts?

Group – B

Answer any four questions of the following :

3. In an examination of 675 candidates ; the examiner supplied the following data :

Class Interval	No. of Candidates
0 – 10	7
10 – 20	32
20 – 30	56
30 – 40	106
40 – 50	180
50 – 60	164
60 – 70	86
70 – 80	44
80 – 90	0
90 – 100	0
Total	675

Find the mean, median and mode of the marks obtained.

OR

Briefly explain using suitable formulae the following :

- (a) Measures of Central Tendency
- (b) Measure of Variation
- (c) Measure of Skewness

4. Find Karl Pearson Coefficient of Correlation from the following data :

x	y
100	30
200	50
300	60
400	80
500	100
600	110
700	130

Explain the value obtained.

OR

Explain briefly the concept of correlation. What is Coefficient of Correlation? Explain the types of Correlation. What is Scatter Diagram? Explain Coefficient of Determination.

5. A husband and wife appear in an interview for two vacancies in the same post. The probability of selection of the husband is $\frac{1}{7}$ and the probability of selection of the wife is $\frac{1}{5}$. What is the probability that :

- (a) Both of them will be selected
- (b) Only one of them will be selected
- (c) None of them will be selected

OR

Explain Addition Law and Multiplication Law of Probability. Explain Bayes' Theorem using the concepts of Prior Probability and Posterior Probability.

6. Explain the concept of Index Numbers. What are its applications? Explain the difference between Unweighted Index Numbers and Weighted Index Numbers. What is consumer Price Index Number?

OR

Explain the concept of Point Estimation and Interval Estimation. What is meant by Confidence Level? Explain the difference between Probabilistic Sampling Methods and Non-probabilistic Sampling Methods. What is meant by Sampling Distribution of the Mean?

7. In an anti-malaria campaign in a certain area, quinine was administered to 812 persons out of a total population of 3248. The number of fever cases reported is shown below :

Treatment	Fever	No Fever	Total
Quinine	20	792	812
No Quinine	220	2216	2436
Total	240	3008	3248

Using Chi-Square Test ; prove whether quinine is useful in checking malaria or not.

OR

Explain the concept of Business Forecasting. What are the different methods of Forecasting?

8. (a) Using Matrix Method, solve the following system of linear equations :

$$3x + 4y = 10$$

$$7x + 11y = 21$$

(b) Differentiate between Matrices and Determinants.

(c) Multiply the following Matrices :

$$\begin{bmatrix} 3 & 7 & 2 \\ 11 & 14 & 17 \\ 5 & 4 & 10 \end{bmatrix} \text{ and } \begin{bmatrix} 11 & 15 \\ 16 & 4 \\ 8 & 5 \end{bmatrix}$$